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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Fifth Semester B.Tech (Minor in CSE) (CBCS) Degree Examinations

January 2025

CS2104-1 – OVERVIEW OF OPERATING SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is an operating system?**
A) interface between the hardware and application programs
B) collection of programs that manages hardware resources
C) system service provider to the application programs
D) all of the mentioned
2. **In Operating Systems, which of the following is/are CPU scheduling algorithms?**
A) Priority
B) Round Robin
C) Shortest Job First
D) All of the mentioned
3. **To access the services of the operating system, the interface is provided by the**
A) Library
B) System calls
C) Assembly instructions
D) API
4. **When a process is in a "Blocked" state waiting for some I/O service. When the service is completed, it goes to the**
A) Terminated state
B) Suspended state
C) Running state
D) Ready state
5. **For an effective operating system, when to check for deadlock?**
A) every time a resource request is made at fixed time intervals
B) at fixed time intervals
C) every time a resource request is made
D) none of the mentioned
6. **Where are placed the list of processes that are prepared to be executed and waiting?**
A) Job queue
B) Ready queue
C) Execution queue
D) Process queue
7. **The interval from the time of submission of a process to the time of completion is termed as**
A) waiting time
B) turnaround time
C) response time
D) throughput
8. **Which scheduling algorithm allocates the CPU first to the process that requests the CPU first?**
A) first-come, first-served scheduling
B) shortest job scheduling
C) priority scheduling
D) none of the mentioned
9. **In priority scheduling algorithm** _____
A) CPU is allocated to the process with highest priority
B) CPU is allocated to the process with lowest priority
C) Equal priority processes can not be scheduled
D) None of the mentioned
10. **Process are classified into different groups in** _____
A) shortest job scheduling algorithm
B) round robin scheduling algorithm
C) priority scheduling algorithm
D) multilevel queue scheduling algorithm
11. **In multilevel feedback scheduling algorithm** _____
A) a process can move to a different classified ready queue
B) classification of ready queue is permanent
C) processes are not classified into groups
D) none of the mentioned

12. Which of the following condition is required for a deadlock to be possible?
 A) mutual exclusion B) a process may hold allocated resources while awaiting assignment of other resources
 C) no resource can be forcibly removed from a process holding it D) all of the mentioned
13. A system is in the safe state if _____
 A) the system can allocate resources to each process in some order and still avoid a deadlock B) there exist a safe sequence
 C) all of the mentioned D) none of the mentioned
14. To avoid deadlock _____
 A) there must be a fixed number of resources to allocate B) resource allocation must be done only once
 C) all deadlocked processes must be aborted D) inversion technique can be used
15. Concurrent access to shared data may result in _____
 A) data consistency B) data insecurity
 C) data inconsistency D) none of the mentioned
16. The segment of code in which the process may change common variables, update tables, write into files is known as _____
 A) program B) critical section
 C) non – critical section D) synchronizing
17. Mutual exclusion implies that _____
 A) if a process is executing in its critical section, then no other process must be executing in their critical sections B) if a process is executing in its critical section, then other processes must be executing in their critical sections
 C) if a process is executing in its critical section, then all the resources of the system must be blocked until it finishes execution D) none of the mentioned
18. Because of virtual memory, the memory can be shared among _____
 A) processes B) threads
 C) instructions D) none of the mentioned
19. The pager concerns with the _____
 A) individual page of a process B) entire process
 C) entire thread D) first page of a process
20. In FIFO page replacement algorithm, when a page must be replaced _____
 A) oldest page is chosen B) newest page is chosen
 C) random page is chosen D) none of the mentioned

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | | | | | | | |
|---|--------------|--------------|------------|-----|----|---|---|--|----|---|---|--|----|---|---|--|----|---|---|--|---|----|---|---|
| 1. a) Define operating systems. Explain the dual mode of operating systems with the help of a neat diagram. | 6 | L*2 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |
| b) Illustrate the different states of a process with the help of a neat diagram. | 6 | L2 | 2 | 1 | | | | | | | | | | | | | | | | | | | | |
| c) Discuss different multithreading models with the help of neat diagrams. | 4 | L2 | 2 | 1 | | | | | | | | | | | | | | | | | | | | |
| 2. a) Discuss the services offered by the operating system. | 5 | L2 | 1 | 2 | | | | | | | | | | | | | | | | | | | | |
| b) Explain two models of IPC (Inter-Process Communication) with the help of a diagram. | 6 | L2 | 2 | 1 | | | | | | | | | | | | | | | | | | | | |
| c) Calculate the average waiting time and average turnaround time by drawing the Gantt chart using the FCFS (First In First Out) scheduling algorithm. | | | | | | | | | | | | | | | | | | | | | | | | |
| <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 10px;"> <thead> <tr> <th style="width: 25%;">Process</th> <th style="width: 25%;">Arrival Time</th> <th style="width: 25%;">Burst Time</th> <th style="width: 25%;"></th> </tr> </thead> <tbody> <tr> <td>P1</td> <td>0</td> <td>2</td> <td></td> </tr> <tr> <td>P2</td> <td>1</td> <td>2</td> <td></td> </tr> <tr> <td>P3</td> <td>5</td> <td>3</td> <td></td> </tr> <tr> <td>P4</td> <td>6</td> <td>4</td> <td></td> </tr> </tbody> </table> | Process | Arrival Time | Burst Time | | P1 | 0 | 2 | | P2 | 1 | 2 | | P3 | 5 | 3 | | P4 | 6 | 4 | | 5 | L3 | 2 | 3 |
| Process | Arrival Time | Burst Time | | | | | | | | | | | | | | | | | | | | | | |
| P1 | 0 | 2 | | | | | | | | | | | | | | | | | | | | | | |
| P2 | 1 | 2 | | | | | | | | | | | | | | | | | | | | | | |
| P3 | 5 | 3 | | | | | | | | | | | | | | | | | | | | | | |
| P4 | 6 | 4 | | | | | | | | | | | | | | | | | | | | | | |
| 3. a) Discuss system calls with an example of handling a user application invoking the open() system call. | 5 | L2 | 1 | 1 | | | | | | | | | | | | | | | | | | | | |

- b) Explain different CPU scheduling criteria.
- c) Calculate the average waiting time and average turnaround time by drawing the Gantt chart using Round Robin scheduling algorithm. (Time Quantum=2ms)

Process	Arrival Time	Burst Time
P1	0	5
P2	1	4
P3	2	2
P4	4	1

5 L2 2 1

6 L3 2 3

Unit – II

4. a) What is process synchronization? Explain the producer-consumer problem with respect to process synchronization and how it can be solved using semaphores.
- b) Illustrate Peterson's solution to critical section problem.
5. a) Define critical section problem. What are the necessary conditions that the solution to critical section problem must satisfy?
- b) Considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t0 following snapshot of the system has been taken:

8 L2 3 2

8 L2 3 3

4 L2 3 3

Process	Allocation	Max	Available
	A B C	A B C	A B C
P ₀	0 1 0	7 5 3	3 3 2
P ₁	2 0 0	3 2 2	
P ₂	3 0 2	9 0 2	
P ₃	2 1 1	2 2 2	
P ₄	0 0 2	4 3 3	

1. What will be the content of the Need matrix?
2. Is the system in a safe state? If Yes, then what is the safe sequence?
3. What will happen if process P1 requests one additional instance of resource type A and two instances of resource type C?

8 L3 3 3

- c) What is fragmentation? Explain its types.

4 L2 4 1

6. a) What is Deadlock? What are the necessary conditions for the Deadlock to occur?
- b) Discuss the structure of inverted page table with a neat diagram.
- c) Explain the concept of swapping of processes from the main memory with the help of a neat diagram.

6 L2 3 2

6 L2 4 1

4 L2 4 1

Unit – III

7. a) What is demand paging? Explain the steps in handling the page faults using appropriate diagram.
- b) Consider the page references 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4-page frame. Find number of page faults and page hits using Optimal Page Replacement algorithms.
8. a) Discuss in detail about various file operations in a file system.
- b) Given the following sequences 82,170,43,140,24,16,190 with the track 50. What is the total disk travelled by the disk arm using FCFS, SSTF algorithms.
- c) With neat diagram, explain Two-level and Tree directory structure.

8 L2 4 1

8 L3 4 3

4 L2 5 1

8 L3 5 3

4 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
